NewsBot Intelligence System Technical Report

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Purpose and scope

I completed this project individually using the supplied ITAI 2373 starter notebook and the BBC News Classification dataset. NewsBot combines text preprocessing, feature extraction, grammatical analysis, sentiment analysis, classification and named entity recognition. The goal is to make news easier to organize and review. The revised evaluation achieved 95.49% held-out accuracy and 95.35% macro F1. These results are promising for a class project. However, the system still needs more testing before it could be used in a real workplace.

Dataset and preparation

The training CSV contained 1,490 labeled articles across five categories: sport (346), business (336), politics (274), entertainment (273) and tech (261). The largest category was about 1.33 times the smallest. The checks on the dataset found no missing values, blank texts or duplicate article IDs, but found 50 repeated article texts. Removing those texts for the revised classification experiment left 1,440 articles.

The BBC file provided ArticleId, Text and Category. I renamed these fields to article_id, content and category to match the notebook. Unlike the template sample, the file did not supply separate title, date or source columns. I removed unavailable title references and set full_text equal to content. This fixed the missing-column errors without adding information that was not in the dataset. Mean article length was approximately 385 whitespace-separated words, with a range of 90 to 3,345.

Text processing and language analysis

The cleaning function lowercased text, removed URLs and email addresses, removed nonletter characters and removed extra spaces. The preprocessing function tokenized the cleaned text, optionally removed stop words, lemmatized tokens and discarded tokens shorter than three characters. I used the processed text for TF-IDF classification and kept the original text for sentiment and other analyses. Removing numbers and punctuation simplifies the vocabulary but can remove information useful for financial reporting or tone.

TF-IDF converted article vocabulary into numerical features. The original classifier used 5,000 TF-IDF features followed by four sentiment values. POS tagging compared grammatical patterns across categories, and dependency parsing examined relationships between words. These results show how the models labeled the text, so they can still contain mistakes. Because the source text was lowercase, recognizing proper nouns was more difficult.

VADER produced compound, positive, neutral and negative sentiment scores. Named entity recognition extracted people, organizations, locations, dates and monetary expressions. The entity relationship analysis connected names that appeared in the same article. This only shows that the names appeared together, not that the people or organizations actually have a relationship. The system labels sentiment as positive, neutral or negative. It does not identify specific emotions like fear or anger.

Implementation and integration

The original comparison trained Multinomial Naive Bayes, Logistic Regression and SVM. The VADER compound value can be negative, but Multinomial Naive Bayes cannot accept negative inputs. A MinMaxScaler inside the Naive Bayes pipeline made its inputs nonnegative and let the scaler learn only from the training portion of each cross-validation split. Missing NLTK resources were resolved by downloading punkt_tab and averaged_perceptron_tagger_eng.

The prediction code originally added seven placeholder values to the TF-IDF vector, although training used only four sentiment values. I replaced the dummy inputs with actual compound, positive, neutral and negative scores and passed full_text into classify_article. After this change, training and prediction both used 5,004 features in the same order. Rerunning the class definition and its initialization allowed the final demonstration to process all three template articles.

Classification results and evaluation design

The original 298-article test split gave 96.64% accuracy for Naive Bayes, 97.32% for Logistic Regression and 95.64% for SVM. These were the first results, but they had a problem: TF-IDF had been fitted before the split. This meant that the test articles had already affected the vocabulary and word weights. The original comparison also used test accuracy to choose the best model. Because of this, the original scores were not enough for a fair final evaluation.

A separate experiment removed repeated article texts and then split the remaining data. It learned the text features using only the training data within each cross-validation split. It compared TF-IDF alone against TF-IDF plus sentiment and log-length features, and selected settings using training macro F1. TF-IDF alone achieved 0.9728 best cross-validation macro F1, compared with 0.9373 for the combined features. Both configurations selected unigrams and C = 2.0. In this experiment, adding sentiment and length features did not help.

The selected TF-IDF-only pipeline achieved 95.49% accuracy and 95.35% macro F1 on 288 held-out articles. Always choosing the most common category would give only 23.61% accuracy. Macro F1 combines precision and recall and gives each category equal weight. This helped me judge performance across all five categories, not just count correct predictions. Rounded class F1 scores were business 0.96, entertainment 0.97, politics 0.94, sport 0.96 and tech 0.93. Tech recall was 0.89, the lowest among the five categories.

This experiment used a better evaluation process, but the training and test articles still came from the same dataset. This split had also been seen during the project. Testing on a new set of articles would give stronger evidence of how well the model works. The revised TF-IDF-only model is a separate experiment; the integrated demonstrations still reflect the original working pipeline. Their probability scores came from the original model, not the revised one.

Language analysis findings

Business had the highest mean numerical-tag proportion at 3.47%, which makes sense because business articles often discuss money and other figures. Politics had the highest proportion under the specified VB + VBD + VBG measure, approximately 11.26%; sport was about 10.40%. The notebook suggested sport might have the highest value, but the result did not support that idea. Entertainment had the highest mean VADER compound score, 0.5991, while politics had the lowest, 0.0578. These scores alone do not tell us what the public thinks or prove that a news category is biased.

Demonstration and error analysis

The integrated system successfully processed three fictional template articles. The Microsoft acquisition example was classified as tech with 37.42% model probability. This category made sense, but the low score showed uncertainty because acquisitions can also be business stories. The Lakers example was classified as sport with 69.03% probability. The climate example was classified as politics with only 25.97% probability. This showed a limitation: the model had to choose one of five categories even though none was a clear match.

The climate example also received a slightly positive VADER compound score of approximately +0.103 despite its alarming wording. This showed me why I needed to read the article instead of relying only on its sentiment score. The entity output incorrectly grouped "Microsoft Acquires" as an organization. Although the saved NER analysis extracted 42,031 mentions, the total counts predicted spans, not verified entities or unique people and organizations. Successful execution therefore does not mean that every prediction is correct.

Applications and limitations

A newsroom could use category suggestions to route incoming articles to editors. An analyst could use entity extraction to locate articles mentioning an organization, while a library could improve browsing of a news collection. I think the system is most useful for organizing and finding articles. A person should still check uncertain results and any results that could affect someone. The probability scores are estimates, not guarantees that a prediction is correct.

The dataset covers one news source and only five labels. These results do not show how well it would work with other publishers, languages, newer topics or writing styles. Removing exact duplicates helped, but similar stories might still appear in both the training and test sets. Cleaning can discard meaningful numbers, lowercase text can impair name recognition, and automated sentiment can misread context. Wrong categories or names could confuse readers or incorrectly connect someone to an event.

Improvements and individual contribution

Next, I would test the system on new articles, look more closely at wrong category predictions and check whether it extracts names correctly. I would also work on handling topics outside the five categories. Future experiments could compare contextual sentiment methods, transformer representations and topic modeling. I would compare each change with the simpler TF-IDF model to see whether it actually helps. Before using it in a real application, I would also need to check incoming data, monitor mistakes and decide which results need human review.

I carried out the individual project workflow, including dataset preparation, notebook execution, debugging, output review and documentation. The work builds on the course starter notebook. AI assistance supported debugging and supplied additional Student Task analysis code and draft explanations. I used and adapted this support while working through the notebook; I did not write every part from scratch.

Sources and project materials

Sources: ITAI 2373 starter notebook and saved outputs in Midterm_NewsBot_Intelligence_System_EMMANUEL.ipynb; BBC News Classification dataset, https://www.kaggle.com/competitions/learn-ai-bbc/data. The notebook uses NLTK/VADER, spaCy, scikit-learn, pandas, NumPy and Matplotlib. All reported results come from its saved outputs.

Project repository: https://github.com/NjehAbabio/ITAI2373-NewsBot-Midterm